

Hadoop Installation Guide

Hadoop Installation on Ubuntu WSL (Single Node Setup)

PRE-Requisite -Ubuntu Wsl must be installed first .

Step 1: Update Ubuntu

```
sudo apt-get update
```

Step 2: Install Java

```
sudo apt-get install default-jdk
```

Verify Java

```
java -version
```

Step 3: Install SSH

Install SSH Server

```
sudo apt-get install openssh-server
```

Start SSH Service

```
sudo service ssh start
```

Generate SSH Key

```
ssh-keygen -t rsa -P ""
```

👉 Press Enter for all prompts

Enable Passwordless SSH

```
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
```

```
chmod 600 ~/.ssh/authorized_keys
```

```
chmod 700 ~/.ssh
```

Test SSH

```
ssh localhost
```

👉 Type:

```
yes
```

Exit SSH

```
exit
```



Step 4: Download Hadoop 2.6.0

```
wget https://archive.apache.org/dist/hadoop/common/hadoop-2.6.0/hadoop-2.6.0.tar.gz
```

Step 5: Extract Hadoop

```
tar -xvzf hadoop-2.6.0.tar.gz
```

Step 6: Move Hadoop

```
sudo mv hadoop-2.6.0 /usr/local/hadoop
```

```
sudo chown -R adi:adi /usr/local/hadoop
```

Step 7: Find Java Path

```
readlink -f $(which java)
```

Expected:

```
/usr/lib/jvm/java-11-openjdk-amd64/bin/java
```

JAVA_HOME:

```
/usr/lib/jvm/java-11-openjdk-amd64
```

Step 8: Configure .bashrc

```
nano ~/.bashrc
```

Add this at the end:

```
# HADOOP VARIABLES START
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export HADOOP_INSTALL=/usr/local/hadoop
export PATH=$PATH:$HADOOP_INSTALL/bin
export PATH=$PATH:$HADOOP_INSTALL/sbin
export HADOOP_MAPRED_HOME=$HADOOP_INSTALL
export HADOOP_COMMON_HOME=$HADOOP_INSTALL
export HADOOP_HDFS_HOME=$HADOOP_INSTALL
export YARN_HOME=$HADOOP_INSTALL
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_INSTALL/lib/native
export HADOOP_OPTS="-Djava.library.path=$HADOOP_INSTALL/lib"
# HADOOP VARIABLES END
```

Save Nano

- Ctrl + O
- Enter
- Ctrl + X

Apply Changes

```
source ~/.bashrc
```

Verify

```
echo $JAVA_HOME
```

```
echo $HADOOP_INSTALL
```

Step 9: Configure Hadoop

9.1 Configure hadoop-env.sh

```
nano /usr/local/hadoop/etc/hadoop/hadoop-env.sh
```

Find:

```
export JAVA_HOME=${JAVA_HOME}
```

Replace with:

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
```

Save and Exit.

9.2 Configure core-site.xml

```
nano /usr/local/hadoop/etc/hadoop/core-site.xml
```

Replace content inside `<configuration>` with:

```
<configuration>
<property>
  <name>fs.default.name</name>
  <value>hdfs://localhost:9000</value>
</property>
</configuration>
```

9.3 Create Hadoop Directories

```
sudo mkdir -p /usr/local/hadoop_store/hdfs/namenode
```

```
sudo mkdir -p /usr/local/hadoop_store/hdfs/datanode
```

```
sudo chown -R adi:adi /usr/local/hadoop_store
```

9.4 Configure hdfs-site.xml

```
nano /usr/local/hadoop/etc/hadoop/hdfs-site.xml
```

Replace content inside `<configuration>` with:

```
<configuration>

<property>
  <name>dfs.replication</name>
  <value>1</value>
</property>

<property>
  <name>dfs.namenode.name.dir</name>
  <value>file:/usr/local/hadoop_store/hdfs/namenode</value>
</property>

<property>
  <name>dfs.datanode.data.dir</name>
  <value>file:/usr/local/hadoop_store/hdfs/datanode</value>
</property>
```

```
</configuration>
```

9.5 Configure mapred-site.xml

```
cp /usr/local/hadoop/etc/hadoop/mapred-site.xml.template /usr/local/hadoop/etc/hadoop/mapred-site.xml
```

```
nano /usr/local/hadoop/etc/hadoop/mapred-site.xml
```

Replace content inside `<configuration>` with:

```
<configuration>
<property>
  <name>mapreduce.framework.name</name>
  <value>yarn</value>
</property>
</configuration>
```

9.6 Configure yarn-site.xml

```
nano /usr/local/hadoop/etc/hadoop/yarn-site.xml
```

Replace content inside `<configuration>` with:

```
<configuration>

<property>
  <name>yarn.nodemanager.aux-services</name>
  <value>mapreduce_shuffle</value>
</property>
```

```
<property>
  <name>yarn.nodemanager.aux-services.mapreduce.shuffle.class
</name>
  <value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>

</configuration>
```



Step 10: Format Hadoop

```
hdfs namenode -format
```

Expected:

```
Storage directory ... has been successfully formatted.
```



Step 11: Start Hadoop

```
start-all.sh
```

```
start-yarn.sh
```



Step 12: Verify Hadoop

```
jps
```

Expected:

NameNode
DataNode
SecondaryNameNode
ResourceManager
NodeManager

Step 13: Test HDFS Commands

Create Directory

```
hdfs dfs -mkdir /test
```

Create Local File

```
echo "Hello Hadoop" > file.txt
```

Upload File to HDFS

```
hdfs dfs -put file.txt /test
```

List Files

```
hdfs dfs -ls /test
```

Read File

```
hdfs dfs -cat /test/file.txt
```

Expected Output:

Hello Hadoop

Important Notes for WSL

Start SSH Every Time After Opening WSL

```
sudo service ssh start
```

Warnings Like These Can Be Ignored

```
Illegal reflective access
```

```
Unable to load native-hadoop library
```

👉 They are warnings, not errors.

Final Result

If these work:

```
jps
```

and

```
hdfs dfs -cat /test/file.txt
```

then Hadoop installation is successful.

HIVE SETUP

Hive Installation on Ubuntu WSL (After Hadoop Setup)

Prerequisite

Before installing Hive, Hadoop must already be running.

Check:

```
jps
```

Expected:

```
NameNode  
DataNode  
SecondaryNameNode  
ResourceManager  
NodeManager
```

If not running:

```
start-all.sh  
start-yarn.sh
```

Step 1: Download Hive

```
wget https://archive.apache.org/dist/hive/hive-2.3.9/apache-hive-2.3.9-bin.tar.gz
```

Step 2: Extract Hive

```
tar -xvzf apache-hive-2.3.9-bin.tar.gz
```

Step 3: Move Hive

```
sudo mv apache-hive-2.3.9-bin /usr/local/hive
```

```
sudo chown -R adi:adi /usr/local/hive
```

Step 4: Configure Hive Environment Variables

Open `.bashrc`

```
nano ~/.bashrc
```

Add these lines at the end:

```
# HIVE VARIABLES START
export HIVE_HOME=/usr/local/hive
export PATH=$PATH:$HIVE_HOME/bin
# HIVE VARIABLES END
```

Save:

- Ctrl + O
- Enter
- Ctrl + X

Apply changes:

```
source ~/.bashrc
```

Step 5: Verify Hive

```
hive --version
```

Expected:

```
Hive 2.3.9
```

Step 6: Configure Hive

Go to Hive config folder:

```
cd /usr/local/hive/conf
```

Create Hive config file:

```
cp hive-default.xml.template hive-site.xml
```

Step 7: Create Hive Warehouse Directory

```
hdfs dfs -mkdir /tmp
```

```
hdfs dfs -mkdir /user
```

```
hdfs dfs -mkdir /user/hive
```

```
hdfs dfs -mkdir /user/hive/warehouse
```

Step 8: Give Permissions

```
hdfs dfs -chmod g+w /tmp
```

```
hdfs dfs -chmod g+w /user/hive/warehouse
```



Step 9: Initialize Derby Database

Run:

```
schematool -dbType derby -initSchema
```

Expected ending:

```
schemaTool completed
```



Step 10: Start Hive

```
hive
```

Expected:

```
hive>
```



Step 11: Test Hive Queries

Create Database

```
CREATE DATABASE college;
```

Show Databases

```
SHOW DATABASES;
```

Use Database

```
USE college;
```

Create Table

```
CREATE TABLE students (  
  id INT,  
  name STRING  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ',';
```

Insert Data

```
INSERT INTO students VALUES (1, 'Adi');
```

Display Table Data

```
SELECT * FROM students;
```

Expected:

```
1  Adi
```



Exit Hive

```
exit;
```



Important Notes

1. Hadoop must be running before Hive

Always check:

```
jps
```

2. Ignore These Warnings

```
Illegal reflective access
```

```
Unable to load native-hadoop library
```

They are warnings, not errors.



Final Result

If this works:

```
SELECT * FROM students;
```

and shows:

```
1  Adi
```


then Hive installation is successful.